NAGALAND LITERACY AND NUMERACY FEST 2026

Micro-Improvement Project (Numeracy Teacher) — Grade 4

Chapter 6 — Animal Shelters: Designing with Fractions

Why this Micro-Improvement Project (MIP)?

Across Nagaland, students already see land, food, and resources being divided every day: a paddy field shared between family members, a plot marked out for a kitchen garden, a biscuit broken between siblings. This Micro-Improvement Project (MIP) builds on that everyday sense of 'sharing equally' and gives it a name: fractions. Students use the familiar, caring context of building a shelter for stray animals to discover, name, and use the fractions ½ (half), ¼ (one-fourth), and ¾ (three-fourths), and to understand when two fractions are equivalent.

Skill in Focus: Fractions (Parts of a Whole, Half, One-Fourth, Three-Fourths & Equivalent Fractions)
Duration: 1 Week

Learning Outcome

- Identifies that a fraction represents a part of a whole.
- Identifies half, one-fourth, and three-fourths of a whole, and their corresponding symbols 1/2, 1/4, and 3/4.
- Explains the meaning of 1/2, 1/4, and 3/4 (what the numerator and denominator represent).
- Represents 1/2, 1/4, and 3/4 using drawings or shaded parts.
- Explains the concept of equivalent fractions and identifies fractions equivalent to a given fraction.

Key Concepts

- Demonstrates the concept of a fraction as an equal part of a whole, using shapes and grids.
- Explains half, one-fourth, and three-fourths, naming the numerator and denominator in each.
- Represents 1/2, 1/4, and 3/4 through drawings, shaded regions, and a divided shelter-plot grid.
- Identifies equivalent fractions by comparing how many boxes of a whole each fraction occupies.

Competency Addressed (PARAKH 2024 Gaps)

Gap 1 — C-1.3, Fractions as Parts of a Whole: "Represents, compares, and interprets fractions (1/2, 1/4) as parts of a whole, on a number line, and as divisions of whole numbers."

Gap 2 — Equivalent Fractions: "Identifies fractions that are equivalent to a given fraction, using pictorial or concrete representations."

Detailed Tasks

Task-1	Find Out What Students Already Know About Fractions

Objective	To find out, before any formal teaching, what each student already understands about dividing a whole into equal parts, so that Task-2 onward can be pitched at the right level for the class.
Teacher's Note	Needs no extra class time or materials — only careful watching and listening across 2–3 normal school days. Keep it conversational, not a test; jot quick notes on who answers confidently, who hesitates, and who needs a picture or object. Use these notes to pace the 'I Do' steps in Task-2 and Task-4.
Duration	40 minutes

Materials Needed
None required — only the teacher's notebook and pencil to jot observations.

Activity Steps

1. While sharing tiffin or snacks, ask: 'If we divided this equally among all of us, what fraction would each person get?'
2. Show a paper circle folded in half and ask: 'What part of the whole circle is this piece?'
3. While lining up, ask: 'If I gave you half of this rope and your friend one-fourth, who has more?'
4. Note who answers confidently, who needs a drawing or object, and who needs the question repeated pick 4–5 students to track on a simple checklist (Identifies parts of a whole / Knows 1/2 and 1/4 / Uses fraction words / Needs support).

Concept Built
A baseline picture of what each student already understands about dividing a whole into equal parts, used to calibrate the pace of Task-2 onward.

Task-2	Introduce Fractions Through the Poem and the Shelter Story

Objective	To introduce the meaning of 'a fraction is a part of a whole' using the poem 'The Shelter We Share', and to help groups decide which stray animals their shelter will care for, so later fraction work has a real, meaningful context.
Teacher's Note	Read the poem slowly, twice if needed — once for enjoyment, once pausing at 'half' and 'fourth' to ask what those words might mean. Keep the shelter-criteria discussion brief; the goal today is equal sharing, not the final design. Physically fold or cut the paper circle in front of the class — seeing the whole become parts matters more than hearing it described.
Duration	40 minutes

Materials Needed

A circle drawn on the board, or a paper/cardboard circle (stand-in for a biscuit) • Plain paper and pencil for each group • Chart paper for the 'shelter criteria' to stick on the wall • Scissors (optional) or a ruler to draw fold-lines

Activity Steps

1.Read the poem 'The Shelter We Share' aloud. Ask: 'What did the goat and the piglet ask for?' Discuss stray animals the students have seen near their homes or school and what problems those animals face.

2.Share the shelter criteria on chart paper and stick it on the wall: at least 2 animal types; a separate enclosure for each; at least 2 caring features per enclosure; a store room for food; a path connecting all areas.

3.I DO: Draw a circle (the 'whole') on the board. Fold or cut it in half. Explain: 2 equal parts = the whole is split into 2, each person gets 1 part, so the fraction is 1/2 (numerator = 1, denominator = 2).

4.WE DO: Guide groups to divide their own paper circle into 4 equal parts and shade 1 part. Ask: what fraction is this? (1/4).

5.YOU DO: Ask groups to shade 3 of the 4 parts and name that fraction (3/4), explaining in their own words what the 3 and the 4 mean.

Homework Extension

Each group finds out from family or neighbours what their chosen animals need to live happily (food, shelter, water) — and writes 2–3 points per animal.

Concept Built

A fraction represents equal parts of a whole; naming and shading 1/2, 1/4, and 3/4, and explaining what the numerator and denominator mean.

Task-3	Set Up a Low-Cost Materials Corner for the Shelter Grid

Objective	To arrange simple, locally available materials so that every student can draw, divide, and decorate their shelter plot without needing anything costly or hard to find.
Teacher's Note	Ask students, a day in advance, to bring one item each from the list below — most homes and classrooms already have these. Keep materials in one shared corner so every group can reach them; this also reduces the pressure on any one student to bring everything.
Duration	40 minutes

Materials Needed

Old newspaper, chart paper, or the reverse side of used paper • Pencil, ruler, and eraser to draw the 6×6 grid • Crayons, sketch pens, or coloured pencils • Bottle caps, pebbles, tamarind or bean seeds (as counters/props) • Dried leaves, twigs, or bits of straw (for fencing/thatch details) • Thread or jute string (for paths or borders) • Cardboard from old cartons or soap boxes (optional, for the Culminating Showcase model)

Activity Steps

1. Set out the materials corner and let each group collect what they need.
2. Each group draws (or is given) a 6×6 grid on chart paper using a ruler and pencil, making all 36 boxes equal in size; this is their 'plot of land.'

Homework Extension
Bring one item from the materials list for the shared classroom corner.

Concept Built
Preparing an organised, low-cost, equal-sized 36-box grid gives every student a concrete 'whole' they can later divide fairly using fractions.

Task-4	Divide the Shelter Plot Using 1/2, 1/4 and 3/4, and Check Equivalent Fractions

Objective	To apply 1/2, 1/4, and 3/4 to divide the 6×6 shelter-plot grid into key areas (enclosures, store room, path), and to check whether any of these areas are equivalent to one another building both the part-whole idea and the idea of equivalent fractions.
Teacher's Note	Keep the 'I Do' short and concrete: shade boxes on the board before asking students to shade their own. When checking equivalence, go back to 'how many boxes' rather than the rule alone e.g. 18 boxes out of 36 is 1/2 because 18 is exactly half of 36. Pair weaker and stronger students so every group can explain, not just complete, the task.
Duration	40 minutes

Materials Needed
6×6 grid drawn on chart paper (from Task-3) • Pencil, ruler, crayons • The fraction reference table (drawn on the board or copied onto paper)

Activity Steps

- I DO: On a 6×6 grid (36 boxes), shade 18 boxes for a store room and path. Ask: what fraction is this? Guide students to see 18/36 = 1/2, since 18 is half of 36.
- WE DO: Shade 9 boxes for one enclosure. Ask groups what fraction this is (9/36 = 1/4).
- YOU DO: Each group decides how many boxes to give each key area on their own grid (animal enclosures, store room, path), labels every area, and records the fraction of the plot it occupies in a fraction table.
- Equivalent Fractions Check: Draw a reference table on the board showing what 1/2 (18 boxes), 1/4 (9 boxes), and 3/4 (27 boxes) of the 36-box plot look like. Groups compare each of their own areas against this table and identify any area equivalent to 1/2, 1/4, or 3/4 even if written with different numbers, e.g. 12/24 in a smaller section.
- Optional outdoor extension: take the class to a real fenced or marked-out space in the school compound (garden bed, parking shed, sports store corner). Ask: 'If this whole space were divided like our shelter plan, what fraction would you give to each use?' Students estimate and discuss outdoors, then refine their indoor grid design.

Homework Extension
Worksheet (teacher-prepared): 3–4 grids with some boxes already shaded students write the fraction shown, and circle any that equal 1/2, 1/4, or 3/4.

Concept Built
Dividing a 36-box whole into 1/2, 1/4, and 3/4 sections, and recognising equivalent fractions by comparing how many boxes each fraction occupies (e.g., 18/36 = 1/2).

Culminating Showcase

Showcase	Shelter Showcase — Build, Exhibit, and Reflect
Objective	To bring together every concept from Tasks 1–4 by turning the fraction-based shelter plan into a physical model or a detailed sketch, presenting it using fraction language, and reflecting on what was learned about parts of a whole and equivalent fractions.
Teacher's Note	Divide model building roles within each group so every student has a hands-on part that also supports the Socio-Emotional Learning outcome of taking others into account while deciding. A clearly labelled, coloured sketch on chart paper is just as valid as a 3D model if materials or time are limited. Insist that every group states the fraction (1/2, 1/4, 3/4, or an equivalent) for each area during their presentation.
Duration	40 minutes

Materials Needed
Cardboard from old cartons, chart paper, and plain paper • Crayons, sketch pens, glue/paste (homemade from flour if needed) • Twigs, matchsticks, or ice-cream sticks (fencing) • Dried grass, cotton, or leaves (thatch/roofing)
- Bottle caps, clay, or folded-paper cut-outs (animal figures) • Thread or string (paths, borders)

Activity Steps

- Day 1 (15 min): Groups divide model-building roles based on chart paper, enclosure walls, the store room, and simple animal figures. Build using the materials corner set up in Task-3.
- Day 2 (10 min): Presentations each group presents which animals their shelter is for and why; how they divided the plot using 1/2, 1/4, and 3/4; any equivalent fractions they found; and one feature they are proud of.
- Day 2 (10 min): Whole-class reflection What did you learn about dividing a whole into equal parts? How did fractions help you plan fairly? What was one challenge, and how did your group solve it?
- Invite family members, other teachers, or another class to view the classroom exhibition.

Concept Built
Independent application of fractions, equivalence, and part-whole reasoning in one integrated, creative design task built, explained, and exhibited with pride.

By the end of these tasks, teachers and students should have moved from everyday sharing to confidently naming, representing, and applying 1/2, 1/4, 3/4, and equivalent fractions culminating in an original, locally-grounded shelter design that each group can showcase with pride.

Micro-Improvement Cycle of the Numeracy Teacher

In this MIP you focused on:

- Helping students identify that a fraction represents a part of a whole.
- Helping students identify half, one-fourth, and three-fourths of a whole, and their symbols 1/2, 1/4, and 3/4.
- Helping students explain the meaning of 1/2, 1/4, and 3/4 in their own words.
- Supporting students to represent 1/2, 1/4, and 3/4 using drawings or shaded parts.
- Helping students explain the concept of, and identify, equivalent fractions.
- Engaging students in designing and building a shelter model, applying fractions to a real, hands-on task.
- Using a familiar, caring context, an animal shelter and an original poem to connect fractions to everyday life and empathy for animals.

Reflect on Students' Understanding

Observe 5–6 students each week during the tasks above. No written test is needed to watch what students do with circles, grids, and shelter plans, and listen to how they explain their thinking.

Student Name	Identifies Parts of a Whole	Knows 1/2, 1/4, 3/4	Explains the Meaning	Identifies Equivalent Fractions	Needs Support
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐

Scale (for each column): ☐ Not Yet ☐ Sometimes ☐ Consistently

Picture Guide